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GROUP REPORT

Multi-access Edge Computing (MEC); EdgeNative-X: OpenCAPIF Integration with MEC

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Contents

Intellectual Property Rights 4
Foreword 4
Modal verbs terminology 4
Executive summary 4
Introduction 5
1 Scope 6
2 References 6
2.1 Normative references 6
2.2 Informative references 6
3 Definition of terms, symbols and abbreviations 6
3.1 Terms 6
3.2 Symbols 7
3.3 Abbreviations 7
4 ETSI MEC Sandbox integration with openCAPIF (OCF) 7
4.1 Introduction 7
4.2 Onboarding/offboarding of the MEC services 8
4.2.1 Automatic onboarding/offboarding of the MEC services onto OCF Platform 8
4.2.1.0 Overview 8
4.2.1.1 Onboarding flow 8
4.2.1.2 Offboarding flow 9
4.2.1.3 Flow diagram 10
4.2.1.4 Configuration 11
4.2.2 Onboarding/offboarding of services produced by MEC applications 11
4.2.2.0 Overview 11
4.2.2.1 Service registration and CAPIF publishing 11
4.2.2.2 Service deregistration and CAPIF unpublishing 12
4.2.2.3 Flow diagram 13
4.2.3 Conclusion 13
4.3 MEC application access to OCF platform exposed APIs 14
4.3.0 Overview 14
4.3.1 OCF platform as a discoverable MEC service 14
4.3.2 MEC application as API invoker 14
4.3.3 Flow diagram 16
Annex A: Change history 17
History 18

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Foreword

This Group Report (GR) has been produced by ETSI Industry Specification Group (ISG) Multi-access Edge Computing (MEC).

Modal verbs terminology

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Executive summary

The present document describes the integration of the Open CAPIF (OCF) [i.6] with the ETSI MEC Sandbox [i.1]. The integration covers three aspects:

- **Automatic onboarding/offboarding:** when a MEC Sandbox network scenario is activated, the MEC 011 service registers as an API provider on the OCF platform and publishes all MEC platform services. On deactivation, all services are unpublished and the provider is offboarded.
- **Onboarding/offboarding of services produced by MEC applications:** when a service-producing MEC application registers or deregisters a service via the MEC 011 Service Management API (clause 5.2.4 of ETSI GS MEC 011 [i.5]), the service is published to or unpublished from the OCF platform.
- **MEC application access to OCF exposed APIs:** the OCF is itself registered as a discoverable MEC service, enabling MEC applications to discover the OCF endpoints through the standard MEC 011 service discovery mechanism and subsequently act as API invokers.

The integration uses the CAPIF provider roles defined in ETSI TS 123 222 [i.2] (API Management Function, API Exposing Function, API Publishing Function) with mutual TLS authentication and certificate-based security as specified in ETSI TS 133 122 [i.4].

Introduction

The ETSI MEC Sandbox [i.1] is a cloud-hosted environment that allows developers to experiment with ETSI MEC APIs. The Open CAPIF (OCF) [i.6] is an open-source implementation of the Common API Framework (CAPIF) defined by ETSI TS 123 222 [i.2] and ETSI TS 129 222 [i.3].

Integration of OCF platform with the ETSI MEC Sandbox enables MEC platform services to be discoverable via the CAPIF service discovery mechanism, extending the reach of MEC service APIs to third-party consumers that use the CAPIF ecosystem. It also allows MEC applications running on the Sandbox to discover and consume third-party APIs exposed through the OCF platform.

Clause 4 of the present document describes the design and implementation of this integration within the MEC Sandbox Application Enablement service (ETSI GS MEC 011 [i.5]).

1 Scope

The present document covers integration of Open CAPIF platform [i.6] into the ETSI MEC Sandbox [i.1].

2 References

2.1 Normative references

Normative references are not applicable in the present document.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or nonspecific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding but are not required for conformance to the present document.

[i.1] ETSI MEC Sandbox: "Software Architecture".

[i.2] ETSI TS 123 222: "LTE; 5G; Common API Framework for 3GPP Northbound APIs (3GPP TS 23.222 version 18.6.0 Release 18)".

[i.3] ETSI TS 129 222: "LTE; 5G; Common API Framework for 3GPP Northbound APIs (3GPP TS 29.222 version 18.7.0 Release 18)".

[i.4] ETSI TS 133 122: "LTE; 5G; Security aspects of Common API Framework (CAPIF) for 3GPP northbound APIs (3GPP TS 33.122 version 18.4.0 Release 18)".

[i.5] ETSI MEC 011: "Multi-access Edge Computing (MEC); Edge Platform Application Enablement Edge Platform Application Enablement".

[i.6] ETSI SDG OpenCAPIF: "OpenCAPIF".

[i.7] ETSI GR MEC 001: "Multi-access Edge Computing (MEC); Terminology".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI GR MEC 001 [i.7] and the following apply:

API Exposing Function (AEF): CAPIF provider function responsible for exposing service APIs to authorized API invokers

NOTE: As defined in ETSI TS 123 222 [i.2].

API Invoker: entity that discovers and consumes service APIs published on the CAPIF Core Function

NOTE: As defined in ETSI TS 123 222 [i.2].

API Management Function (AMF): CAPIF provider function responsible for managing the lifecycle of the API provider domain registration

NOTE: As defined in ETSI TS 123 222 [i.2].

API Publishing Function (APF): CAPIF provider function responsible for publishing service API information to the CAPIF Core Function

NOTE: As defined in ETSI TS 123 222 [i.2].

CAPIF Core Function (CCF): central component of the CAPIF architecture that provides API registration, discovery, and management capabilities

NOTE: As defined in ETSI TS 123 222 [i.2].

Provider Domain: grouping of CAPIF provider functions (AMF, AEF, APF) that together represent a service provider's presence on the CAPIF platform

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI GR MEC 001 [i.7] and the following apply:

AEF	API Exposing Function
AMF	API Management Function
APF	API Publishing Function
API	Application Programming Interface
CSR	Certificate Signing Request
mTLS	Mutual Transport Layer Security
OCF	Open CAPIF
UUID	Universally Unique Identifier

4 ETSI MEC Sandbox integration with openCAPIF (OCF)

4.1 Introduction

This clause describes the integration of the ETSI MEC Sandbox [i.1] with the OCF platform[i.6]. The integration is implemented within the MEC 011 Application Enablement service [i.5] through a dedicated CAPIF client module (`capif-client`) that manages the full lifecycle of CAPIF provider interactions. It covers the following three aspects:

- **Onboarding/offboarding of MEC Platform Services** of MEC platform services (clause 4.2.1): when the MEC 011 service starts, it automatically registers as an API provider on the OCF platform, generating certificate signing requests for three CAPIF provider functions - API Management Function (AMF), API Exposing Function (AEF), and API Publishing Function (APF) - and obtaining signed certificates for mutual TLS authentication.

- **Onboarding/offboarding of services produced by MEC applications** (clause 4.2.2): when a service-producing MEC application registers or deregisters a service through the MEC 011 (clause 5.2.4 of ETSI GS MEC 011 [i.5]), the service is asynchronously published to or unpublished from the OCF using the APF mutual TLS client.

- **MEC application access to external APIs exposed by OCF platform** (clause 4.3): the OCF platform is registered as a discoverable MEC 011 service, enabling MEC applications to discover OCF platform endpoints through the standard MEC 011 service query mechanism and to act as API invokers on the OCF.

The security model follows ETSI TS 133 122 [i.4], for CCF, and uses:

- **Basic Authentication** for login and user management on the OCF Register server;
- **Bearer tokens** for provider onboarding and offboarding operations;
- **Mutual TLS (mTLS)** with per-role certificates (APF, AMF) for API publishing and provider lifecycle management;
- **OAuth** as the declared security method for published service APIs.

4.2 Onboarding/offboarding of the MEC services

4.2.1 Automatic onboarding/offboarding of the MEC services onto OCF Platform

4.2.1.0 Overview

The goal is to automatically onboard/offboard the MEC platform services [i.1] onto OCF platform, if CAPIF is enabled in the scenario, when a network scenario is activated/deactivated. The MEC services are exposed by the network scenario with a unique API provider domain and associated provider function identifiers (AMF ID, AEF ID, APF ID). The onboarding process is triggered by the MEC 011 service during its initialization phase.

4.2.1.1 Onboarding flow

The onboarding is performed by the CAPIF client module as a background operation during MEC 011 service startup. It uses an exponential backoff retry strategy (up to 6 retries, starting at 10 seconds) to handle transient failures during OCF platform availability. The flow consists of four sequential steps:

Step 1 - Login to OCF Register server

The CAPIF client authenticates with the OCF platform Register server using HTTP Basic Authentication with administrator credentials. The Register server returns an admin access token and a refresh token:

- Method: `POST /login`
- Authentication: Basic Auth (CAPIF_REGISTER_USERNAME : CAPIF_REGISTER_PASSWORD)
- Response: `{access_token, refresh_token}`

Step 2 - Create provider user

Using the admin access token, the CAPIF client creates a new provider user with a UUID-based username. This user will be associated with the MEC Sandbox provider domain:

- Method: `POST /createUser`
- Authentication: Bearer token (admin access token)
- Request body: `{username: <UUID>, password, enterprise: "ETSI", country: "France", email: "mec-sandbox@etsi.org"}`
- Response: `{uuid}`

Step 3 - Obtain authentication materials

The CAPIF client retrieves the authentication materials needed for provider onboarding. This includes an access token, the CA root certificate for TLS verification, and the URLs for all CAPIF Core Function endpoints:

- Method: `GET /getauth`
- Authentication: Basic Auth (provider user credentials)

- Response: `{access_token, ca_root, ccf_api_onboarding_url, ccf_publish_url, ccf_discover_url, ccf_onboarding_url, ccf_security_url}`

Step 4 - Onboard provider with CSRs

The CAPIF client generates three RSA Certificate Signing Requests (CSRs), one for each CAPIF provider function role:

- API Management Function (**AMF**): used for provider lifecycle management (offboarding);
- API Exposing Function (**AEF**): used for exposing APIs to invokers;
- API Publishing Function (**APF**): used for publishing and unpublishing service APIs.

The CSRs are submitted to the CAPIF Core Function, which returns signed certificates and unique function identifiers for each role.

- Method: `POST /{ccf_api_onboarding_url}`
- Authentication: Bearer token (access token from Step 3)
- Request body: `{apiProvFuncs: [{role: AMF, csr}, {role: AEF, csr}, {role: APF, csr}], apiProvDomInfo: <provider_domain>}`
- Response: `{apiProvDomId, apiProvFuncs: [{amfId, cert}, {aefId, cert}, {apfId, cert}]}`

Upon successful onboarding, two mutual TLS HTTP clients are constructed using the signed certificates and the CA root certificate:

- **APF TLS client**: used for all service publishing and unpublishing operations;
- **AMF TLS client**: used for provider offboarding during shutdown.

4.2.1.2 Offboarding flow

The offboarding is triggered when the MEC 011 service shuts down (scenario deactivation or service termination). It consists of three steps:

1) **Unpublish all services:** iterates over all CAPIF service mappings stored in Redis (`capif:svc:*`) and concurrently sends a DELETE request for each published API using the APF TLS client. Up to 10 DELETE requests are issued in parallel, and the entire step is bound by a 25-second timeout to respect the Kubernetes® pod termination grace period. Any service not unpublished within the deadline is skipped with a warning.
2) **Offboard provider:** sends a DELETE request to `/{ccf_api_onboarding_url}/{apiProvDomId}` using the AMF TLS client, removing the provider domain from the OCF.
3) **Delete user:** sends a DELETE request to `/deleteUser/{userUUID}` on the Register server using the admin token, cleaning up the provider user account.

4.2.1.3 Flow diagram

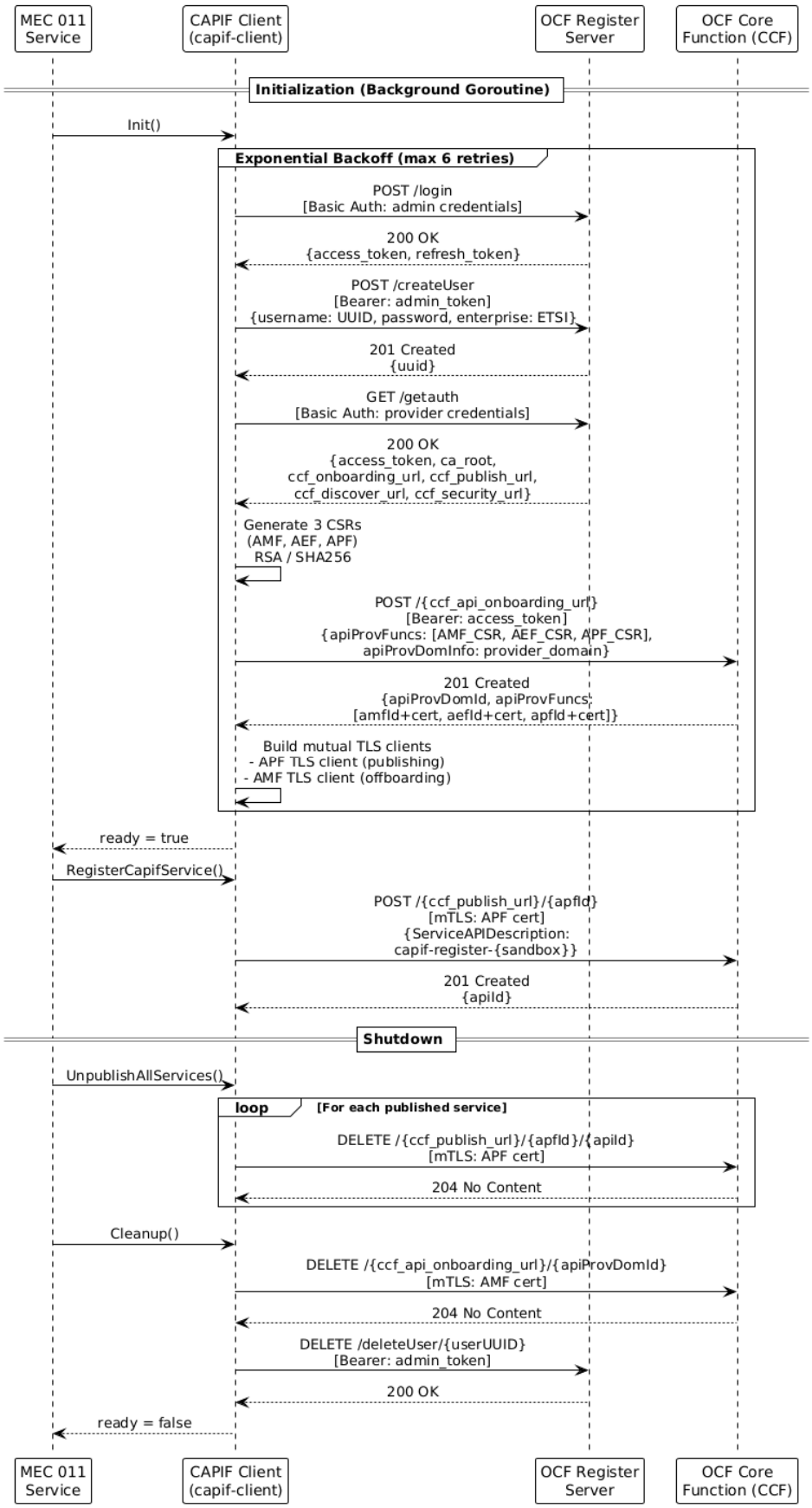


Figure 4.2.1.3-1: Automatic Onboarding/Offboarding Flow

4.2.1.4 Configuration

The following environment variables control the CAPIF integration.

Table 4.2.1.4-1: Environment variables for CAPIF integration

Variable	Description	Default
`CAPIF_ENABLED`	Enable/disable CAPIF integration	`false`
`CAPIF_REGISTER_HOSTNAME`	OCF Register server hostname	-
`CAPIF_REGISTER_PORT`	OCF Register server port	`443`
`CAPIF_REGISTER_USERNAME`	Administrator login username	-
`CAPIF_REGISTER_PASSWORD`	Administrator login password	-
`CAPIF_USER_PASSWORD`	Provider user password	-
`CAPIF_HOSTNAME`	CAPIF Core Function hostname	-
`CAPIF_PORT`	CAPIF Core Function port	`443`
`CAPIF_PROVIDER_DOMAIN`	Provider domain identifier	`MECSandbox_to_CAPIF_Provider`

4.2.2 Onboarding/offboarding of services produced by MEC applications

4.2.2.0 Overview

The purpose of the OCF onboarding/offboarding of services produced by MEC applications is to expose the MEC service APIs on the OCF when a service-producing MEC application registers a service, and to remove them when the service is deregistered.

The onboarding/offboarding is triggered by the MEC 011 Service Management API (clause 5.2.4 of ETSI GS MEC 011 [i.5]). On adding a new MEC service, the MEC 011 service triggers the publishing of the service to the OCF. On removing a MEC service, the MEC 011 service triggers the unpublishing of the service from the OCF.

4.2.2.1 Service registration and CAPIF publishing

When a MEC application registers a new service via `POST /applications/{appInstanceId}/services`, the MEC 011 service:

1) Validates the request (ensuring the application state is READY and required fields are present).
2) Generates a unique service instance identifier (`serInstanceId`).
3) Stores the `ServiceInfo` in Redis.
4) Sends a `ServiceAvailabilityNotification` with change type `ADDED` to subscribed applications.
5) Returns HTTP 201 Created with the `ServiceInfo` and a `Location` header.
6) **Asynchronously** (in a separate goroutine) publishes the service to the OCF.

The asynchronous CAPIF publishing step constructs a `ServiceAPIDescription` as defined in ETSI TS 129 222 [i.3] with the following properties.

Table 4.2.2.1-1: ServiceAPIDescription properties

Property	Value
API name	`{meep_platform}-{serName}-{serCategory}-{sandboxName}`
AEF ID	Obtained during automatic onboarding (clause 4.2.1)
Protocol	`HTTP_1_1`
Data format	`JSON`
Security methods	`OAUTH`
Resource operations	`GET`, `POST`, `PUT`, `DELETE`
Resource URI	`/{sandbox_name}/{meep_mep_name}/{MEC_SERVICE}/{version}/`
Interface address	MEC platform hostname and port

The publishing uses the **APF mutual TLS client** established during onboarding. Upon successful publication (HTTP 201), the OCF returns an `apiId` which is stored as a mapping in Redis (`capif:svc:{serInstanceId}` → `apiId`) for later retrieval during unpublishing.

If the initial publish attempt fails (network error, OCF unavailable, or session expiry), the failure is not silently discarded. Instead, a **retry strategy** is applied:

- The failed operation has persisted to Redis under the key `capif:pending:pub:{serInstanceId}`, recording the full publish parameters, retry count, and last error message.
- A background worker (`runCapifRetryWorker`) polls for pending operations every 30 seconds and re-attempts each one while the CAPIF client is ready.
- After 5 consecutive failed retries, the entry is discarded and an error is logged, preventing unbounded retry storms.

A 401 Unauthorized response from the OCF (session/certificate expiry) additionally triggers an automatic re-initialization of the CAPIF session (new user, new certificates) before the retry, handled transparently by `PublishServiceAPI`.

If a stale CAPIF mapping is found in Redis when publishing (e.g. from a previous pod lifecycle), it is proactively unpublished first to avoid 403 duplicate-name conflicts.

NOTE: The asynchronous pattern prevents the CAPIF publishing from blocking the MEC 011 API response to the MEC application.

4.2.2.2 Service deregistration and CAPIF unpublishing

When a MEC application deregisters a service via `DELETE /applications/{appInstanceId}/services/{serviceId}`, the MEC 011 service:

1) Retrieves and deletes the `ServiceInfo` from Redis.
2) Sends a `ServiceAvailabilityNotification` with change type `REMOVED` to subscribed applications.
3) Returns HTTP 204 No Content.
4) **Asynchronously** unpublishes the service from the OCF:
   a) Retrieves the CAPIF `apiId` from the Redis mapping.
   b) Sends `DELETE /{ccf_publish_url}/{apfId}/{apiId}` using the APF mutual TLS client.
   c) On success (HTTP 204), deletes the Redis mapping.
   d) If the unpublish fails, the operation has persisted to Redis under `capif:pending:unpub:{serInstanceId}` and retried by the background worker every 30 seconds, up to 5 times. On exhaustion, the entry is discarded and the Redis mapping is cleaned up. A 404 response (API already absent) is treated as a non-error success.

4.2.2.3 Flow diagram

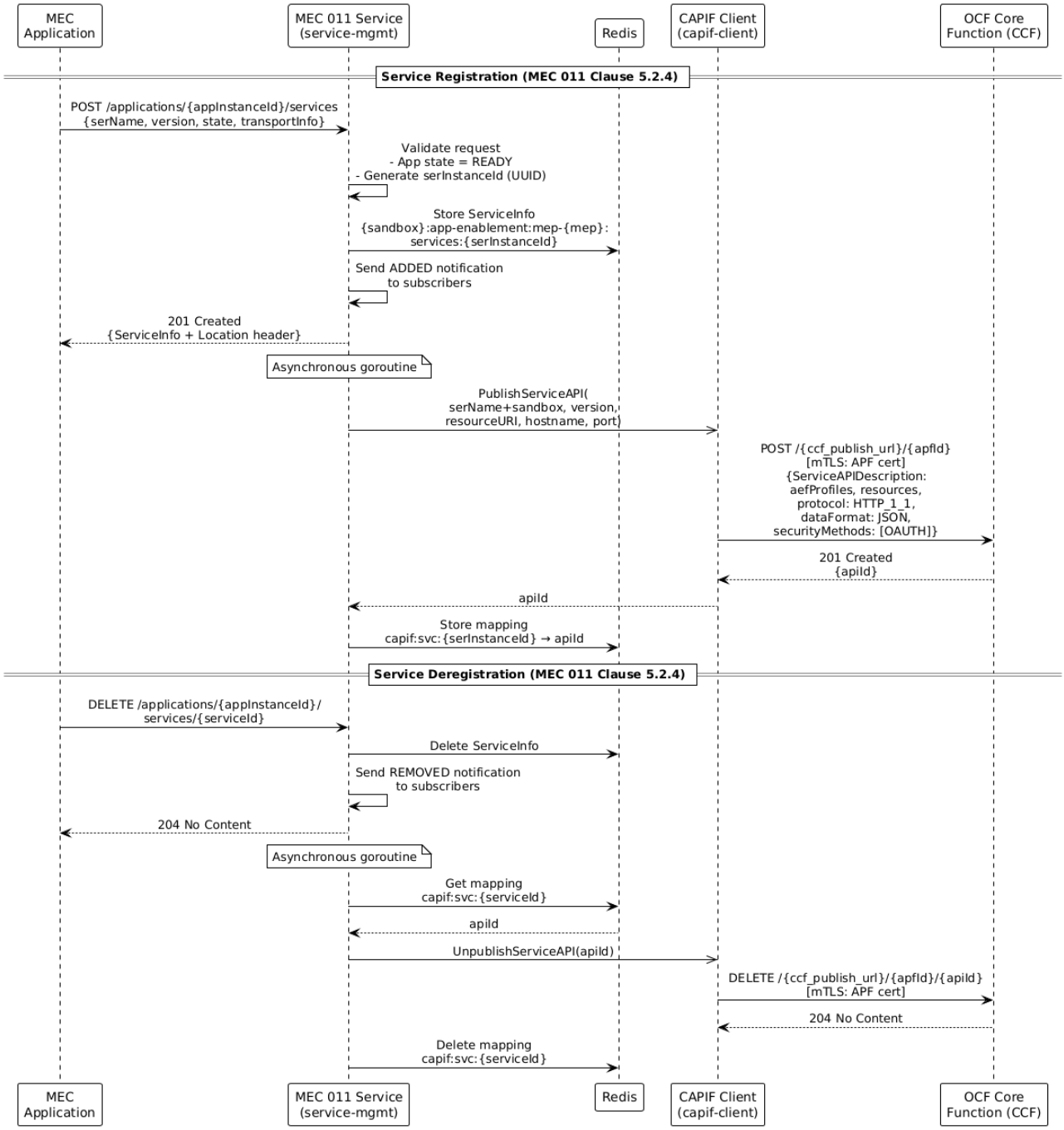


Figure 4.2.2.3-1: Onboarding/Offboarding of Services Produced by MEC Applications

4.2.3 Conclusion

The implementation uses a single `capif-client` module within the MEC 011 service that provides a common API for both automatic and MEC application-produced service publishing:

- The **automatic onboarding** (clause 4.2.1) uses `capif-client.Init()` to establish the provider session (login, user creation, certificate exchange) and `capif-client.Cleanup()` for teardown.
- The **service-producing MEC application onboarding** (clause 4.2.2) uses `capif-client.PublishServiceAPI()` and `capif-client.UnpublishServiceAPI()` - the same methods used internally for automatic publishing.

Two resilience mechanisms are implemented to ensure consistency between MEC 011 service state and OCF state:

- **Retry strategy:** failed asynchronous publish and unpublish operations have persisted in Redis and retried by a background worker every 30 seconds for up to 5 attempts. Operations that exceed the retry limit are logged and discarded. Session expiry (HTTP 401) triggers automatic CAPIF re-initialization before the retry.
- **Concurrent graceful shutdown:** during offboarding, all published services are unpublished concurrently (up to 10 parallel requests) within a 25-second deadline, ensuring the pod terminates within the Kubernetes® grace period.

4.3 MEC application access to OCF platform exposed APIs

4.3.0 Overview

The objective is to offer MEC applications the ability to access the third-party APIs exposed by the OCF platform. The solution exposes the OCF as a MEC service, discoverable by MEC application using MEC 011 service APIs. MEC application can use the provided endpoints to connect to the OCF.

4.3.1 OCF platform as a discoverable MEC service

Once the CAPIF client is successfully onboarded (clause 4.2.1), the MEC 011 service registers OCF platform as a discoverable service with the following properties.

Table 4.3.1-1: OCF platform as a discoverable service properties

Property	Value
Service name	`capif-register-{sandboxName}`
Category	`CAPIF` (version v3)
Service version	`3.0.0`
State	`ACTIVE`
Transport type	`REST_HTTP`
Protocol	`HTTP`
Serializer	`JSON`
Scope of locality	`MEC_SYSTEM`

The `TransportInfo` includes five endpoint URIs that expose the full OCF registration and discovery workflow.

Table 4.3.1-2: Endpoints for the OCF registration and discovery

Endpoint	Port	Purpose
`https://{host}:{registerPort}/login`	Register port	Administrator authentication
`https://{host}:{registerPort}/createUser`	Register port	User account creation
`https://{host}:{registerPort}/getauth`	Register port	Provider/invoker authentication material retrieval
`https://{host}:{capifPort}/discover`	CAPIF port	API discovery
`https://{host}:{capifPort}/onboard`	CAPIF port	Provider/invoker onboarding

This service is also published to the OCF itself (using `PublishServiceAPI`), enabling discovery from both MEC 011 and CAPIF perspectives.

4.3.2 MEC application as API invoker

A MEC application can act as an API invoker on the OCF by following this workflow:

Step 1 - Discover the OCF platform via MEC 011

The MEC application queries the MEC 011 Service Management API to discover available services:

- Method: `GET /mec_service_mgmt/v1/services?ser_category_id=capifId`

- Response: `ServiceInfo` for `capif-register-{sandbox}` including endpoint URIs.

Step 2 - Register as an API invoker

Using the OCF endpoints obtained in Step 1, the MEC application:

1) Authenticates with the Register server (`POST /login`).
2) Creates an invoker user account (`POST /createUser`).
3) Obtains authentication materials (`GET /getauth`), including the CA root certificate and CCF endpoint URLs.
4) Generates a CSR for the API Invoker function.
5) Onboards as an API invoker on the CAPIF Core Function (`POST /{ccf_onboarding_url}`), submitting the CSR and receiving a signed certificate and `apiInvokerId`.

Step 3 - Discover and invoke published APIs

The MEC application can now discover all APIs published on the OCF:

- Method: `GET /{ccf_discover_url}?api-invoker-id={apiInvokerId}` (using invoker mutual TLS)
- Response: `ServiceAPIDescriptions` containing the API names, AEF profiles, resource URIs, and interface descriptions (IPv4 addresses and ports).

The MEC application extracts the endpoint information from the `ServiceAPIDescription` and invokes the discovered MEC services directly.

Step 4 - Cleanup

On termination, the MEC application offboards from the OCF (`DELETE /{ccf_onboarding_url}/{apiInvokerId}`) and deletes its user account.

4.3.3 Flow diagram

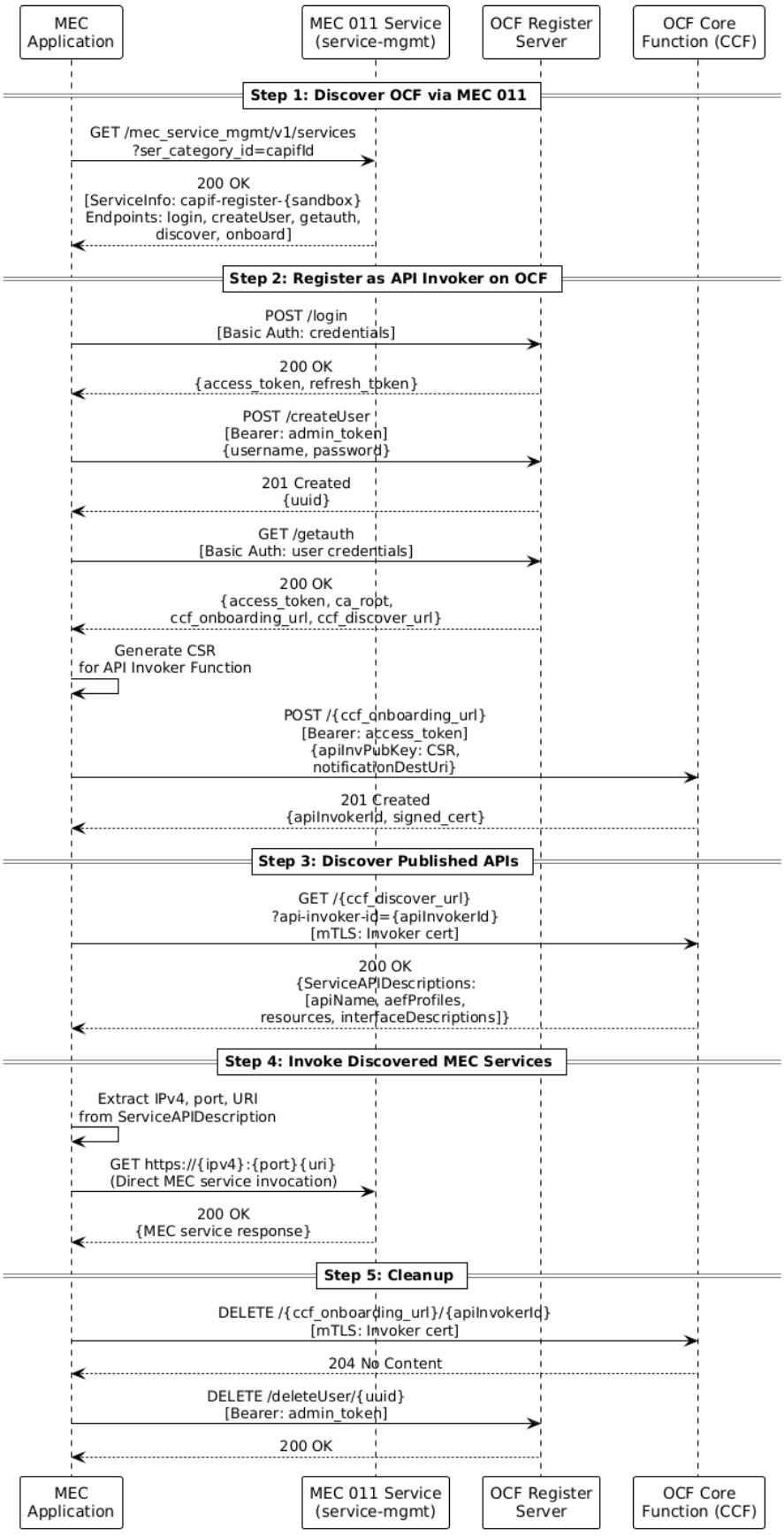


Figure 4.3.3-1: MEC Application Access to OCF Exposed APIs

Annex A:
Change history

Date	Version	Information about changes
March 2026	V0.0.1	Early draft: document skeleton with clause structure.
June 2026	V4.0.1	Changes to prepare stable draft: • Populated Clause 4 (OCF integration) based on capif-client implementation. • Added terms, abbreviations, executive summary, introduction. • Added PlantUML flow diagrams. • Final improvements and report ready to be stable.

History

Version	Date	Status
V4.1.1	September 2026	Publication